

```

#include <stdio.h>
#define MAX 10
int main() {
    int n;
    int burst[MAX], deadline[MAX], period[MAX];
    int remaining[MAX];
    int time = 0;
    printf("Enter number of tasks: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        printf("Task %d (burst deadline period): ", i + 1);
        scanf("%d %d %d", &burst[i], &deadline[i], &period[i]);
        remaining[i] = 0; // initially no job released
    }
    printf("\nPID\tBurst\tDeadline\tPeriod\n");
    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t%d\t%d\n", i + 1, burst[i], deadline[i], period[i]);
    }
    int limit = 20;
    printf("\nScheduling occurs for %d ms\n\n", limit);

    while (time < limit) {
        for (int i = 0; i < n; i++) {
            if (time % period[i] == 0) {
                remaining[i] = burst[i];
            }
        }
        int min_deadline = 9999;
        int selected = -1;
        for (int i = 0; i < n; i++) {
            if (remaining[i] > 0) {
                if (deadline[i] < min_deadline) {
                    min_deadline = deadline[i];
                    selected = i;
                }
            }
        }
        if (selected == -1) {
            printf("%dms : CPU is idle.\n", time);
        } else {
            printf("%dms : Task %d is running.\n", time, selected + 1);
            remaining[selected]--;
        }
        time++;
    }
}

```

```
}  
return 0;  
}
```

```
C:\Users\Admin\Desktop\1 x + v  
Enter number of tasks: 3  
Task 1 (burst deadline period): 2  
4  
5  
Task 2 (burst deadline period): 3  
7  
20  
Task 3 (burst deadline period): 2  
8  
10  
  
PID    Burst  Deadline  Period  
1       2      4         5  
2       3      7        20  
3       2      8        10  
  
Scheduling occurs for 20 ms  
  
0ms : Task 1 is running.  
1ms : Task 1 is running.  
2ms : Task 2 is running.  
3ms : Task 2 is running.  
4ms : Task 2 is running.  
5ms : Task 1 is running.  
6ms : Task 1 is running.  
7ms : Task 3 is running.  
8ms : Task 3 is running.  
9ms : CPU is idle.  
10ms : Task 1 is running.  
11ms : Task 1 is running.  
12ms : Task 3 is running.  
13ms : Task 3 is running.  
14ms : CPU is idle.  
15ms : Task 1 is running.  
16ms : Task 1 is running.
```

1WA24CS232